

ROOTS-AI™ Phase 1 MVP Master Product & Implementation Requirements v3.3.2

FINAL CONTROLLED INTEGRITY UPDATE — Appendix C replacement

This v3.3.2 update supersedes the Appendix C filenames/versions/hashes and the prior v3.3.1 correction-sheet Appendix C verification rows only. All other v3.3.1 substantive corrections and all unaffected v3.3 requirements remain unchanged. This is a version-control/integrity update; it does not alter question wording/order, scoring logic, milestones, commercial scope or acceptance thresholds.

Annex	Final controlled source	Version	SHA-256
C-01	02_ROOTS_AI_C01_Canonical_Question_Bank_v1.0.1_CORRECTED.xlsx	v1.0.1	b5bb50ded8a21ec957d17f55c9b2b03c4845ad56d850549cb997bff4bfdd354b
C-02	03_ROOTS_AI_C02_Canonical_Scoring_Rules_and_Golden_Tests_v1.0.1_CORRECTED.xlsx	v1.0.1	14bf61735d0abe42da7ea87c5dd988dcceecdd648b94a68137b18a65f978989e
C-03	04_ROOTS_AI_C03_Report_Content_and_Visual_Reference_Pack_v1.0.1_CORRECTED.pdf	v1.0.1	04d9ab36fac863352ccc04580cbf7d225331ea7249cdae02f1ef06b09565e5ef
C-04	05_ROOTS_AI_C04_Website_Content_and_Legal_Copy_Pack_v1.0.1_CORRECTED.docx	v1.0.1	4866c20d12266a42f3325dc5212bbb571ddb91a597d1a72e669e438ffbdbbb550
C-05	07_ROOTS_AI_C05_UI_UX_Screen_Implementation_Specification_v1.0.1_CORRECTED.pdf	v1.0.1	3fbc330a848cb640cdf6f5d703ec3f83297d0f5e6296ddd52e31781bfb0c98d0

Visual authority: the final visual-design process is governed by File 14 v1.0.1 CORRECTED FINAL and the written-approved PUB-01 Golden Screen in ROOTS-owned editable Figma. File 15 v1.0.1 controls PUB-01 functional/content/interaction requirements. File 16 v1.3.3 contains functional/content wireframes only. Legacy G02/V01 or SVG filenames do not override this authority.

Precedence: if any attached legacy Appendix C row or prior correction-sheet row shows v1.0.0 or an earlier filename/hash, this v3.3.2 page controls.

ROOTS-AI™ Phase 1 MVP

Master Product, Technical & Acceptance Requirements — Version 3.3

Document ID: ROOTS-MVP-MASTER-REQ-006

Status: APPROVED CONTROLLED IMPLEMENTATION BASELINE

Effective date: upon internal approval by ROOTS AI HEALTH SYSTEMS, Inc.

Prepared exclusively for: ROOTS AI HEALTH SYSTEMS, Inc.

MASTER BASELINE. This document consolidates the complete technical, functional, security, delivery, evidence and acceptance requirements for the Phase 1 MVP. It is the sole controlled source of truth for product planning, procurement, implementation, verification and acceptance. Only requirements expressly stated here and in its controlled data annexes form the approved baseline.

Confidential — Proprietary intellectual property of ROOTS AI HEALTH SYSTEMS, Inc.

Document Control

Field	Binding value
Document owner	ROOTS AI HEALTH SYSTEMS, Inc. (Delaware, USA)
Product	ROOTS-AI™ Phase 1 Minimum Viable Product
Version	3.3
Classification	Confidential — Controlled Implementation Baseline
Intended use	Product planning, supplier selection, implementation, verification, UAT and final acceptance
Approval authority	ROOTS AI HEALTH SYSTEMS, Inc.
Change authority	Formally approved ROOTS change record; no change by implication, silence or oral discussion

Revision History

Version	Date	Status	Purpose
1.0	July 2026	Superseded	Initial binding scope-lock review
2.0	July 2026	Superseded	Contract-oriented master specification
3.0	July 2026	Approved controlled implementation baseline	Supplier-neutral consolidated product, technical and acceptance requirements
3.1	21 July 2026	Approved controlled release	Published executable C-01 through C-04 annexes; removed all content-release blockers
3.2	21 July 2026	Approved controlled release	Compressed delivery baseline to 30 calendar days including seven-day UAT; aligned commercial execution release
3.3	21 July 2026	Approved controlled release	Added binding C-05 page-by-page UI/UX, navigation, role, responsive and screen-state specification

Sole-Reference Governance

This Master Requirements document eliminates parallel technical interpretations. The hierarchy below is exhaustive for Phase 1 product and implementation decisions.

Priority	Document	Version / status	Role
1	This Master Product & Implementation Requirements document and its Controlled Data Annexes	v3.3 — approved controlled release	Sole control of product scope, behavior, delivery, evidence and acceptance

Priority	Document	Version / status	Role
2	ROOTS-approved change records	Approved after Effective Date	Modify only the expressly identified requirements
3	Executed Software Development Agreement	Executed with selected implementation partner	Commercial and legal terms; must incorporate this technical baseline without reducing it

Conflict and Interpretation Rules

- For all product scope, functional behavior, content implementation, security, delivery evidence and acceptance, this Master Requirements document controls.
- The Controlled Data Annexes listed in Appendix C are components of this document, not separate interpretive specifications. Their recorded version and SHA-256 hash identify the only executable data release.
- No implementation team may infer question mappings, rules, formulas or content from historical diagrams, emails, samples or unapproved working papers.
- A more detailed requirement does not constitute scope expansion when it explains how a required function must operate.
- Any function required for a listed deliverable to operate safely, securely and as specified is included even if it is an enabling component rather than a visible screen.
- Work needed to correct non-conformance is a defect remedy, not a Change Request.

1. Purpose, Product Boundary and Binding Scope

The Phase 1 MVP is a web-based biological assessment and reporting platform. It collects a structured 73-question assessment across 13 modules, applies deterministic rules to calculate seven domain scores and derived measures, identifies probable biological drivers, generates a governed personalized narrative, assembles a 19-section report, and supports controlled administration and de-identified research export.

The MVP is an educational biological intelligence and wellness product. It is not a diagnostic device, does not replace a licensed healthcare professional, does not diagnose disease, does not prescribe treatment, and does not autonomously make clinical decisions.

1.1 In-Scope Operational Capabilities

- Responsive public website and legal pages defined in Section 6.
- Passwordless authentication limited to assessment resume, report access and administration.
- Versioned 73-question, 13-module assessment with validation, progress, auto-save, resume and controlled missing-value behavior.
- Deterministic seven-domain scoring and approved derived measures.
- Rules-based driver, confidence and evidence logic.
- Governed LLM narrative layer restricted to language generation.
- Interactive web report and pixel-consistent downloadable PDF containing 19 required sections.
- Admin Dashboard for participant operations, report operations, configuration control, audit review and research export.
- Development, staging and production environments; automated deployment pipeline; logging, backup and monitoring baseline.
- All testing, documentation, source code, credentials and knowledge transfer required for independent operation by ROOTS-AI™.

1.2 Explicitly Out of Scope

Excluded capability	Phase 1 treatment
Laboratory result ingestion or biomarker analysis	Not built; public content may describe it as Coming Soon
DNA, epigenetic or nutrigenomic analysis	Not built; Coming Soon only
Microbiome analysis	Not built; Coming Soon only
Wearable API integrations	No Apple Health, Garmin, Fitbit, Oura, WHOOP or real-time streaming
Clinician portal or patient management	No operational clinical portal; informational page only
Longitudinal user dashboard	No ongoing health tracking dashboard; resume and report access only

Excluded capability	Phase 1 treatment
EHR/FHIR/HL7 integration	Deferred
Conversational AI coach	Deferred
Native iOS or Android application	Web application only
Multilingual interface	English only
Marketplace, billing or subscriptions	Deferred
Enterprise or population-health analytics	Deferred
Formal SOC 2, HITRUST, HIPAA certification or penetration-test certification	Deferred; core architecture and controls remain mandatory
Full offline application	Excluded; transient same-device queueing for connection loss is resilience, not an offline product

1.3 Non-Negotiable Product Boundary

The implementation team shall not silently add, remove, substitute, rename or materially simplify any in-scope capability. Optional enhancements may be proposed, but they shall not displace a required deliverable, consume an agreed acceptance milestone, or alter the approved budget or timeline unless authorized through a formal ROOTS change record.

2. Target Users, Roles and Access Model

Role	Permitted access	Prohibited access
Participant	Own assessment draft, own submitted answers, own generated report	Any other participant, scoring configuration, audit data or research export
Admin	Participant operations, reports, approved configuration functions	Direct database superuser credentials; silent alteration of published scoring rules
Research Admin	Approved de-identified export only	Names, emails, direct identifiers, reports or raw authentication data
Super Admin	Role assignment, system configuration and audit review	Bypassing audit logging or deleting audit history
Implementation Team	Time-bound least-privilege access required for delivery/support	Ownership of production accounts; secondary use of data; permanent access after handover

2.1 Age and Consent

- The technical input range is 16–110 years to preserve the approved questionnaire design.
- Production launch shall default to age 18+ unless ROOTS-AI™ supplies an approved jurisdiction-specific minor-consent workflow.
- If 16–17-year-old participation is enabled, consent, guardian requirements, privacy notice and research eligibility shall be explicitly configured and tested.
- The Recovery Potential age band for an eligible 16–17-year-old shall use the 18–30 band unless this Master Requirements document is changed through a formally approved ROOTS change record.

2.2 Authentication Boundary

- Magic Link authentication is required for secure resume and report access; it does not create a full user dashboard.
- Admin and Super Admin accounts require MFA.
- Links and sessions require configurable expiry, one-time or replay-resistant validation, logout, revocation, rate limiting and audit events.
- Report storage is private. A database identifier or token appearing in a URL shall not itself authorize access.

3. Locked Technical Architecture

Layer	Required implementation
Frontend	Next.js / React with TypeScript strict mode; Tailwind CSS; mobile-first responsive design
API	Server-side TypeScript REST API with versioned endpoints and OpenAPI/Swagger documentation
Scoring	Deterministic configuration-driven service; no LLM involvement in scores, drivers or classifications
Narrative	Server-side OpenAI API integration using governed templates, structured minimized inputs and validated outputs
PDF	Node.js Puppeteer using approved HTML/CSS templates and visual regression references
Data/Auth/Storage	Supabase PostgreSQL, Supabase Auth and private Supabase Storage
Hosting	Vercel for the web/API baseline; a dedicated approved worker may be used for Puppeteer only when documented and approved
CI/CD	GitHub Actions or equivalent in a ROOTS-AI™-owned repository
Environments	Separate Development, Staging and Production with distinct credentials and data

3.1 Architecture Gate

No material coding milestone is accepted until the implementation team supplies and ROOTS-AI™ approves:

- System architecture and data-flow diagrams.
- Final service and hosting inventory with expected recurring costs.
- Database ERD, table list and RLS policy map.
- API inventory and authentication model.
- PHI/PII data-flow map showing every processor and external service.
- Decision record for any deviation from the locked stack.

3.2 Code Quality

- Protected main branch, develop integration branch, reviewed pull requests and standardized commits.
- ESLint and Prettier enforced; TypeScript strict mode with no unexplained any types in business logic.
- Modular, reusable components; business rules are configuration-driven, not scattered through UI code.
- Secrets are stored in environment/secret management and never committed.

- All dependencies are maintained, commercially usable, security-reviewed and recorded in the SBOM.
- Copyleft dependencies require prior written approval.

4. Binding Data Model

The MVP shall implement at least the eight logical application tables below. The implementation team may normalize further, but may not collapse auditability, consent, versioning or access isolation.

Supabase-managed auth tables, views and storage buckets are additional.

Table	Minimum purpose and mandatory attributes
profiles	Participant identity/profile, auth mapping, demographic inputs, status, timestamps; direct identifiers segregated
assessments	One record per assessment; questionnaire_version, status/state, started/completed times, progress, device/browser metadata, retention state
responses	Assessment/question link, raw value, normalized value, is_na, validation state, answer timestamps and source version
scores	Seven domain scores, Biological State, Opportunity, Recovery Potential, Confidence, drivers, evidence, scoring_version and calculation trace reference
reports	Assessment link, report version, status, private storage reference, generation metadata, prompt/rules/model versions and checksum
scoring_config	Versioned questions/mappings/weights/thresholds/templates; draft/published/retired state; effective time; publisher; rollback target
consents	Append-only service, privacy and research consent records; document version, purpose, timestamp, jurisdiction and withdrawal state
audit_logs	Append-only access, authentication, admin, configuration, export, report, AI governance and security events with actor/time/action/object/result

4.1 Required Views and Storage

- research_export_v1: de-identified/pseudonymized approved fields only, restricted to Research Admin and separately audited.
- Private report bucket: no public URLs; time-limited signed access only after authorization.
- No production PHI or participant identifiers in Development or ordinary application logs.

4.2 RLS and Database Security

- RLS enabled and tested on every application table and protected view; no table is exempt merely because it is not listed in an older example.
- Participant access is constrained by verified auth identity and assessment ownership.
- Admin roles are implemented through verified claims/role tables, never by exposing service-role credentials to the browser.
- Every write operation validates authorization on the server and database layers.
- Negative RLS tests must prove that cross-user reads, writes, exports and report retrieval are blocked.

4.3 Research Pseudonymization

The MD5 plus static-salt example in earlier documents is prohibited. Research identifiers shall use HMAC-SHA-256 or an approved equivalent with a secret key stored outside source code. Research export requires explicit research consent, field minimization, export logging and a documented withdrawal process. Pseudonymized data shall not be described as anonymous unless the approved methodology supports that claim.

5. Assessment Experience — 73 Questions / 13 Modules

The exact text, answer options, module order, question IDs, conditional logic and scientific mappings shall be loaded only from the Canonical Question Bank and Scoring Rules datasets recorded in Appendix C. Once their version, SHA-256 hash and internal approval record are entered, those datasets form controlled annexes to this Master Requirements document. No historical questionnaire, draft matrix, sample report or other document may be used.

5.0 Canonical Content Freeze

- The Canonical Question Bank shall contain exactly 73 stable question IDs assigned across exactly 13 ordered modules.
- The Scoring Rules dataset shall identify every scored question, domain mapping, weight, reverse-scoring flag, threshold, formula constant, protective factor and classification boundary.
- Both datasets require schema validation, human-readable export, version number, approval date and SHA-256 hash recorded in Appendix C.
- If either dataset is absent, incomplete, internally inconsistent or does not match its recorded hash, the implementation team shall stop the affected work and raise a written clarification. Guessing or importing an earlier draft is prohibited.
- A corrected content dataset is not a scope change when it resolves a pre-existing ambiguity before the content-freeze gate; after approval, changes follow the formal ROOTS change-control process and create a new version.

5.1 Functional Requirements

- 73 questions across exactly 13 modules, each with stable IDs and a recorded questionnaire version.
- One-question or approved grouped-question presentation according to the Assessment Specification.
- Visible module label and percentage progress calculated from eligible questions.
- Previous/Next navigation without data loss; confirmation before destructive restart.
- Auto-save on each answer with visible Saving/Saved/Failed state and retry.
- Resume after browser close using secure authenticated assessment state.
- Waist cm/in toggle with conversion to canonical cm and preservation of entered unit/value.
- Keyboard operation and shortcuts; touch targets and focus order compliant with WCAG 2.1 AA.
- Loading, empty, timeout, session-expired, conflict, offline-transition, submission and report-generation states.
- Transient same-device queueing during a connection interruption, with safe synchronization and conflict handling after reconnection; no promise of full offline use.

5.2 Validation

Input	Rule	Acceptance behavior
Age	16–110 technical range; launch eligibility rule applies	Block invalid values; display approved error; no silent clamping
Height	100–250 cm	Block invalid values
Weight	25–350 kg	Block invalid values
Waist	40–200 cm after conversion	Block invalid values; conversion precision tested
Scale answer	Only approved option/value pair	Reject forged or out-of-range API input
Conditional question	Required only when activation rule is true	Hidden question does not create a zero or stale score
Free text	Only where expressly approved; length and content sanitation	Never directly concatenated into prompts or queries

5.3 Missing and N/A Rules

- N/A is excluded from both numerator and denominator; it is never zero and never replaced by a default answer.
- No Answer is missing, not N/A, and is never imputed as 2 or any other value.
- If fewer than 50% of eligible questions for a domain are answered, that domain score is null and the report explains that it could not be calculated.
- If overall completion is below the completion threshold defined in the Appendix C rules release, submission is blocked or explicitly flagged according to this Master Requirements document.
- All denominator, null-domain and derived-score behavior must be covered by deterministic tests.

5.4 State Machine

Not Started → Started/Draft → Paused → Started → Submitted → Scoring → Generating Report → Completed → Archived. Error states must be recoverable without duplicate submissions or lost answers.

6. Public Website and Page Inventory

Phase 1 includes the operational and informational pages below. Informational pages about future layers do not create those future capabilities. No CMS is required unless separately approved; content may be implemented as version-controlled static content.

Route / page	Phase 1 requirement
/ — Home	Hero, value proposition, six report features, trust language and assessment CTA
/assessment	Complete assessment workflow
/report/[id]	Authorized interactive 19-section report and PDF download
/example-report	Anonymized approved sample report
/how-it-works	Assessment → deterministic analysis → governed report → educational action
/platform	Assessment and AI Engine marked Available; Labs, DNA, Microbiome, Wearables and Biological Twin marked Coming Soon
/research	Research vision, privacy, ethics, collaboration and pilot information; no operational cohort dashboard
/healthcare-professionals	Informational professional page; no clinician portal or patient list
/pilot	Free Beta/Pilot eligibility, benefits, consent-aware join flow, timeline and FAQs
/about	Mission, science, framework and company information
/contact	Support, research and business inquiries with secure submission/routing
/blog	Blog landing and approved initial content or a professionally designed launch state
/privacy	Privacy notice, data rights, processors, retention and contact channel
/terms	Terms of Service
/cookies	Cookie/analytics notice and consent controls where applicable
/medical-disclaimer	Educational use, non-diagnostic status and emergency limitations
/ai-disclaimer	Role, limitations and governance of AI-generated language
/admin	Protected operational Admin Dashboard

6.1 Design and Content System

- Quiet Luxury visual language: Deep Navy #1A2A4A, Soft Navy #2A4060, Warm White #FAF8F8, Charcoal #1A1A1A and approved state colors.
- Inter typography, mobile-first responsive rules, reusable components and consistent spacing tokens.
- English primary interface; all product names and trademarks use the approved naming conventions.
- Medical-grade, calm, educational and empathetic language; no fear-based or diagnostic copy.
- WCAG 2.1 AA target with keyboard access, labels, contrast, focus visibility and screen-reader semantics.

6.2 Analytics Boundary

Analytics may measure public marketing funnels only after required consent. GA4, Microsoft Clarity, session replay, pixels and equivalent tools are prohibited on assessment, report, authentication and admin routes and shall never receive answers, health scores, identifiers, tokens or report content.

7. Deterministic Scoring and Intelligence Pipeline

The approved pipeline is: 73 Answers → Validation/Normalization → Seven Domain Scores → Biological State™ → Opportunity Score™ → Recovery Potential™ → Drivers/Evidence/Confidence → Governed Narratives → 19-Section Report. Each stage shall be independently testable and versioned.

7.1 Scientific Source of Truth

- The Appendix C Canonical Question Bank and Scoring Rules datasets govern all mappings, reverse-scored questions, weights, thresholds, protective factors and formula constants.
- Historical examples, sample reports, code snippets, diagrams and unapproved working papers are non-authoritative.
- The implementation team may not add age, sex/gender or other demographic adjustments unless they appear in the Appendix C Scoring Rules dataset.
- No missing response may be imputed. No rule may be silently invented to make a report complete.

7.2 Required Outputs

Output	Required properties
Seven domain scores	0–100 or null; exact inputs, denominator, mappings and scoring version traceable
Biological State™	Deterministic derived score; null-domain policy explicit and tested
ROOTS Opportunity Score™	Deterministic formula and interpretation from the Appendix C rules release
ROOTS Recovery Potential™	Multi-factor deterministic formula with factor contributions visible to the audit trace
Drivers	Primary, secondary, tertiary and co-primary logic from the Appendix C rules release; no LLM selection
Evidence Strength	Rules-based evidence category based on approved coverage/consistency logic
ROOTS Confidence™	Rules-based confidence reflecting consistency, answer confidence and missingness
Explainability object	Per-output source question IDs, normalized values, weights, contributions, rule/version and warnings

7.3 Configuration Lifecycle

- Draft → Reviewed → Published → Retired states.
- Only authorized Super Admin users may publish or roll back.
- Every report is permanently linked to questionnaire, scoring, rules, template, prompt and model versions used.
- Published historical versions are immutable; changes create a new version.
- A rollback restores configuration for new reports without altering historical outputs.

7.4 Golden Test Pack

Before UAT, the implementation team shall deliver and pass a ROOTS-AI™-approved Golden Test Pack containing at least 30 deterministic cases covering:

- Every domain, every reverse-scored question and every classification boundary.
- All N/A, partial N/A, below-threshold domain coverage and null-derived-score cases.
- Minimum/maximum numeric inputs and waist conversion.
- Driver ties, co-primary cases, evidence/confidence boundaries and protective-factor combinations.
- Recalculation under a new configuration version without modification of historical results.

8. LLM Narrative Layer and AI Governance

The LLM is a language renderer only. It shall receive structured, minimized, pre-calculated explanation objects and shall never calculate, change or override a score, classification, driver, evidence level, confidence level, warning or action-rule eligibility.

8.1 Mandatory Controls

- Prompts, governance rules and scoring logic are server-side confidential assets.
- Questionnaire free text is never concatenated directly into system or model-instruction prompts.
- Inputs are allow-listed structured fields with identifiers removed or minimized.
- Output must conform to an approved JSON schema, length limits and section-specific template contract.
- Prohibited claims are checked after generation; failure triggers one governed retry and then deterministic fallback content.
- The system shall not diagnose, prescribe, instruct medication cessation, claim certainty, fabricate test results or invent participant facts.
- Conditional language is used where scientifically appropriate; governance shall not rely only on a brittle keyword list.
- Every narrative event records report ID, model/provider, prompt version, rules version, input-object hash, outcome, violations, retry/fallback and timestamp.
- Prompts and outputs are not displayed in ordinary user logs or exposed to participants.

8.2 Permitted and Prohibited Language

Permitted framing	Prohibited framing
May, suggests, possible, potential, associated with, appears consistent with	You have, this proves, definitely, diagnosis, cure, treat, prescribe
Educational options and micro-steps from approved rule libraries	Medication changes, individualized medical treatment, emergency guidance beyond approved escalation
Encouragement and explanation of scores already calculated	New causes, conditions, scores or tests invented by the LLM

8.3 Provider Failure

- OpenAI timeout or error shall never affect deterministic scores.
- The MVP fallback is an approved deterministic template library, not an unapproved Llama/Mistral deployment.
- An alternative model/provider requires a formally approved ROOTS change record, privacy/security review and regression testing.

9. ROOTS Biological Report™ — 19 Required Sections

#	Section	Minimum content
1	Cover Page	Identity, report ID/version, generation date and non-diagnostic label
2	Executive Summary	Governed overview using approved explanation objects
3	ROOTS Biological State™	Score/badge, classification and concise explanation
4	ROOTS Opportunity Score™	Score, interpretation and limits
5	ROOTS Confidence™	Confidence level, contributing data quality and missingness
6	Key Drivers	Primary, secondary, tertiary/co-primary with evidence
7	Seven-Domain Score Breakdown	All available domain scores, classifications and visual bars
8	Biological Triad™	Approved relationship diagram and explanation
9	Future Projection	Conditional educational projection, never a medical prognosis
10	90-Day Roadmap	Approved phased educational roadmap
11	Nutrition Priorities	Rule-library content; no disease treatment claims
12	Action Priorities	Ranked micro-actions with approved rationale
13	What Is Going Well	Protective factors and strengths
14	Specific Concerns	Signals requiring attention, expressed conditionally
15	Suggested Laboratory Discussion	Optional tests to discuss with a clinician; no result interpretation or requirement
16	Participant Answers	Approved response summary with privacy-conscious layout
17	Biological Card	Compact summary of scores, drivers, versions and confidence
18	Final Word	Governed personalized closing
19	Medical and AI Disclaimer	Mandatory limitations, emergency direction and professional-care statement

9.1 Report Rules

- All 19 sections must be present or explicitly show an approved Not Available state; a download button is not a report section.
- Interactive report and PDF use the same canonical report JSON and display the same scores, versions and approved narrative.
- The PDF reference design, page-break rules, fonts, charts, colors and minimum/maximum content cases are approved before acceptance.
- Reports are generated from immutable assessment and scoring snapshots.
- A report-generation retry shall not silently produce different scores or drivers.
- Every PDF carries report ID, generation timestamp, report version and disclaimer.

10. Admin Dashboard and Research Export

10.1 Admin Dashboard

- Participant list with search, filter, status and permitted detail view.
- Assessment status and failed-job review without silent answer alteration.
- Report generation/retry/status management and authorized access to generated reports.
- Scoring configuration draft/review/publish/retire/rollback functions with immutable history.
- Role and permissions management restricted to Super Admin.
- Audit and AI governance log viewer with filters and export for authorized users.
- Research export generation restricted by consent and role.
- Operational metrics: assessment starts/completions, abandonment, report success/failure and downloads, without exposing health data to third-party analytics.

10.2 Research Export

- CSV and JSON export from an approved database view.
- Separate explicit research consent; service consent alone is insufficient.
- No full name, email, authentication identifier, Magic Link token or report narrative.
- Field list, generalization/suppression rules and small-cell controls approved before launch.
- Every export records actor, purpose, query/filter, row count, checksum, time and recipient/project reference.
- Withdrawal process prevents new exports and follows the approved research protocol for existing datasets.

11. Security, Privacy and Compliance Baseline

Control	Mandatory Phase 1 requirement
Encryption	TLS/HTTPS for all endpoints; provider encryption at rest; private report storage; secrets outside source
Identity	Magic Link protections, MFA for privileged roles, least privilege, session revocation and rate limiting
RLS	Enabled and negatively tested on every application table/view
Audit	Access, auth, admin, config, export, report and AI governance events; minimum 12-month retention
Backup	Daily automated database backups, minimum 30-day retention, PITR where supported, documented quarterly restore test
Vulnerability	Dependency scan and OWASP Top 10 application scan before UAT; Critical/High findings resolved
Data minimization	Collect and send only data required for the approved function
Deletion/retention	Documented retention rules and tested user/admin deletion workflow subject to legal/research obligations
Incident notice	Implementation team notifies ROOTS-AI™ within 24 hours of suspected unauthorized access, loss or security event and preserves evidence
Subprocessors	Complete list approved before production; no unapproved subcontractor or processor
Cross-border transfer	Documented locations, processors and applicable safeguards
Production data	No production participant data in Development; sanitized test fixtures only

11.1 PHI and External Providers

- No PHI is stored before ROOTS-AI™ confirms the required Supabase BAA, eligible plan/add-on and configuration.
- If PHI is transmitted to OpenAI or another processor, the appropriate contract/BAA and eligible service configuration must be confirmed before use; otherwise inputs must be de-identified/minimized so PHI is not transmitted.
- The implementation team shall produce a data-flow map and processor register covering Supabase, OpenAI, hosting, email, monitoring and any analytics service.
- Formal HIPAA, SOC 2 or HITRUST certification is not promised by the MVP. Marketing and UI shall use precise language such as designed to support future compliance, never falsely certified or compliant by architecture alone.

12. Performance, Reliability and Compatibility

Metric	Binding target and test condition
Public/assessment page load	≤2 seconds under mutually agreed production-like conditions
Save response	p95 ≤500 ms excluding network conditions outside the service boundary
Next question interaction	≤300 ms after local validation
Non-AI API endpoints	p95 ≤500 ms under agreed representative load
Admin queries	≤3 seconds for datasets up to 100,000 records using agreed filters/indexes
Report generation	≤15 seconds for at least 95% of approved staging test cases; governed fallback used on provider timeout
Report display	≤2 seconds after authorized data retrieval
Availability objective	99.9% monthly during production warranty/support, excluding agreed provider-wide outages and scheduled maintenance

12.1 Supported Clients

- Latest two stable versions of Chrome, Edge, Safari and Firefox on desktop.
- Current and previous major iOS Safari and Android Chrome versions at UAT start.
- Responsive layouts for agreed mobile, tablet and desktop breakpoints.
- Accessibility keyboard and screen-reader smoke tests on critical workflows.

13. Testing and Acceptance Evidence

13.1 Required Test Suites

Suite	Minimum acceptance requirement
Unit	≥80% coverage for business logic and 100% coverage of critical scoring formula branches
Golden scoring	At least 30 approved deterministic cases; exact expected JSON outputs
Integration	Auth, API, database, scoring, narrative, PDF, storage, email and admin integrations
End-to-end	Start/resume/submit/report/download; admin/config/export; error recovery
Security	RLS negative cases, access-control tests, OWASP Top 10 scan, dependency and secret scan
Performance	Evidence against every metric in Section 12
Accessibility	Automated plus manual critical-flow WCAG 2.1 AA checks
Compatibility	Agreed desktop/mobile browser matrix
PDF visual	Reference-based visual regression across short, long, missing-data and boundary cases
AI governance	Adversarial and boundary prompts, prohibited claims, schema failures, retry and fallback
Backup/restore	Backup configuration evidence and one successful pre-launch restore test

13.2 UAT Rules

- The seven-calendar-day UAT period must begin no later than Day 24 after all in-scope functions, Staging deployment, test reports and user instructions are delivered in writing. The Implementation Partner remains responsible for achieving Final Acceptance by Day 30 unless ROOTS approves a written extension.
- ROOTS-AI™ may issue a defect list during UAT. Material remediation pauses the UAT clock; the affected functionality receives a reasonable retest period after redelivery.
- Silence or passage of time is not acceptance.
- Final acceptance requires zero open Critical or High defects. Medium defects require a ROOTS-approved remediation schedule; Low defects require recorded disposition.
- Signing an interim demo, invoice or milestone receipt does not waive later-discovered non-conformance in integrated behavior.

13.3 Definition of Complete

A deliverable is complete only when it is implemented, integrated, tested, documented, deployed to Staging, demonstrated as working software, supported by the required evidence and accepted under UAT. A mockup, slide, isolated code branch, incomplete API, unreviewed document or manual workaround is not complete.

14. Required Documentation and Handover

#	Handover item
1	Source code in ROOTS-AI™-owned repository
2	README and local setup instructions
3	Installation and deployment guides
4	Environment variable register without secret values
5	OpenAPI/Swagger documentation with examples and error codes
6	Database schema, ERD and migration files
7	RLS policy document and negative-test evidence
8	Role and permissions matrix
9	Architecture and end-to-end data-flow diagrams
10	PHI/PII processor and subprocessor data-flow register
11	Encryption, storage and report-access implementation document
12	Backup, retention and restore evidence
13	Audit log specification and samples
14	Scoring configuration schema, published release and Golden Test Pack
15	Prompt/template/governance version register
16	Admin Dashboard user manual
17	Test reports, coverage, OWASP scan, performance and accessibility evidence
18	Bug tracker and UAT completion report
19	SBOM and third-party license register
20	Recurring service/cost inventory and account ownership list
21	Privacy, Terms, Cookie, Medical and AI disclaimer source pages
22	2–4-hour recorded knowledge-transfer session and technical Q&A
23	All production, hosting, domain, email, Supabase, OpenAI and monitoring access transferred/confirmed
24	All C-05 UI/UX handover artifacts, editable assets, component catalog and screen evidence

14.1 Independent Operability

Handover is sufficient only when a competent replacement team can build, deploy, configure, test and operate the MVP without undocumented implementation-team knowledge or a proprietary dependency that was not disclosed and licensed.

15. Project Governance, Milestones and Commercial Controls

15.1 Working Method

- Weekly working-software demonstrations are mandatory; slides alone do not satisfy a demo.
- A shared issue tracker and decision log record requirements, defects, decisions, owners and dates.
- The implementation team identifies dependencies, decisions and blockers early; silence does not transfer responsibility to ROOTS.
- Material subcontracting or production-data access requires prior ROOTS approval and equivalent confidentiality, intellectual-property and data obligations.

15.2 Delivery Schedule

Gate	Target	Required outcome
G0 — Kickoff	Day 0	Executed documents, ROOTS-AI™ repository, accounts, plan and team confirmed
G1 — Architecture	End Day 3	Architecture Gate artifacts approved
G2 — Assessment/Data	End Day 7	Schema/RLS, auth, 73-question assessment, validation and save/resume on Staging
G3 — Intelligence/Report	End Day 14	Deterministic engine, governance, 19-section web/PDF report and secure delivery
G4 — Complete MVP	End Day 20	All pages, Admin, export, monitoring, security and integrated workflows
G5 — Release Candidate	No later than End Day 23	All tests/documents complete; production-like Staging candidate
UAT	Days 24–30	Seven-calendar-day formal UAT after complete delivery
Final Acceptance	No later than Day 30	Production deployment, complete handover, acceptance record and final payment gate

15.3 Payment Protection

Payments are earned by accepted outcomes, not the passage of a week. The Agreement should retain at least 25% of the fixed fee until Final Acceptance. Any invoice for a technical milestone requires the evidence and Client approval defined for that gate. Third-party subscription, hosting, BAA/add-on and API charges must be disclosed in writing before commitment.

15.4 Change Requests

- A Change Request must identify the exact requirement changed, reason, cost, timeline, security/scientific impact and affected tests/documents.
- No changed implementation begins until ROOTS formally approves the change record.
- Bug fixes, missing enabling work, security corrections and compliance with this Master Requirements document are not Change Requests.

16. Intellectual Property, Repository and Accounts

- The Git repository is created under and owned by ROOTS-AI™ from Day 0. The implementation team receives revocable role-based access; repository ownership remains with ROOTS-AI™ at all times.
- All custom code, schemas, UI/UX, prompts, templates, configurations, scoring implementations, documentation, tests and related work product transfer under the Agreement upon payment of the applicable accepted work.
- Prompt engineering, scientific mappings, scoring methodology, Recovery Potential logic and all Phase 1 modifications are confidential ROOTS-AI™ trade secrets and may not be reused or repurposed.
- All pre-existing third-party or implementation-team materials must be disclosed in an approved register before use. Any embedded pre-existing component requires a perpetual, irrevocable, worldwide, transferable, sublicensable, royalty-free license sufficient to operate, modify and commercialize the Deliverables.
- All production accounts, domains, certificates, cloud projects, API organizations and billing accounts shall be owned or controlled by ROOTS-AI™ wherever the provider permits.
- No implementation team, contractor or service provider may use ROOTS-AI™ data, prompts, outputs or Confidential Information to train, evaluate or improve another product or model.

17. Warranty and Complimentary Support

- The 90-day defect warranty and three-calendar-month complimentary support period commence on Final Acceptance and apply concurrently; neither narrows the other.
- Warranty covers implementation defects, specification non-conformance, regression defects and security/configuration errors attributable to delivered work.
- Complimentary support includes minor bugs, reasonable configuration/deployment adjustment, minor non-scope-changing UI refinements and operational technical assistance.
- Third-party outages are excluded only when all required handling, fallback and monitoring were implemented and the outage was not caused or worsened by negligent configuration.

Severity	Acknowledgement target	Restoration/remedy target
Critical	4 business hours	Continuous reasonable effort; workaround/restoration within 1 business day where feasible
High	1 business day	Remedy or accepted workaround within 3 business days
Medium	2 business days	Planned remedy within 10 business days
Low	3 business days	Disposition in the next agreed maintenance cycle

18. Acceptance Criteria — Final Binding Checklist

ID	Acceptance criterion
AC-01	All in-scope pages, workflows and enabling services operate as specified.
AC-02	The Appendix C canonical 73-question, 13-module dataset is implemented verbatim, hash-verified and versioned.
AC-03	All deterministic scores and derived measures pass the approved Golden Test Pack.
AC-04	N/A, missing values, reverse scoring, null domains and boundary behavior pass.
AC-05	LLM cannot alter scores/drivers and passes governance, schema, retry and fallback tests.
AC-06	The interactive and PDF reports contain all 19 required sections and pass visual review.
AC-07	Magic Link, session, report authorization and privileged MFA controls pass.
AC-08	All eight minimum logical tables, protected views and private storage are implemented and documented.
AC-09	RLS and role controls pass positive and negative access tests.
AC-10	Consent, research export, deletion/retention and audit workflows pass.
AC-11	All performance targets are met under agreed test conditions.
AC-12	Daily backup, 30-day retention, applicable PITR and a successful restore are evidenced.
AC-13	OWASP/dependency/security testing has zero open Critical or High findings.
AC-14	Unit, integration, E2E, accessibility, compatibility, PDF and AI governance reports pass.
AC-15	All required public, legal and informational pages are published on Staging.
AC-16	Admin, configuration lifecycle, audit viewer and research export operate under approved roles.
AC-17	Development, Staging and Production environments and CI/CD are established under ROOTS-controlled accounts.
AC-18	All documentation, SBOM, license, processor and cost registers are delivered and reviewed.
AC-19	Repository, credentials, accounts and access are under ROOTS-AI™ control.
AC-20	Recorded knowledge transfer and technical Q&A are complete.
AC-21	No Critical or High defects remain; Medium/Low disposition is formally approved and recorded.
AC-22	Seven-day UAT is completed and the ROOTS Project Acceptance Record is approved.
AC-23	All intellectual-property, confidentiality, conformance and handover evidence is complete and retained.
AC-24	Every C-05 screen, navigation path, role state, responsive breakpoint, error/empty/loading state and UI evidence item passes.

19. Internal Governance and Procurement Use

This Master Requirements document is maintained by ROOTS AI HEALTH SYSTEMS, Inc. It may be issued as a controlled procurement or implementation baseline to any future team without changing its ownership, requirement IDs or acceptance criteria.

Governance rule	Internal requirement
Document ownership	ROOTS AI HEALTH SYSTEMS, Inc. retains ownership and change authority
Controlled issue	Every issued copy records the document version, date and file hash
Supplier comparison	Proposals are compared against Appendix A requirement IDs and evidence expectations
Deviations	Any proposed deviation is assessed and approved or rejected by ROOTS before implementation
Scope protection	A quotation, estimate, assumption or proposal omission cannot reduce the approved baseline
Historical copies	Superseded copies are archived and may not be used for active implementation
Acceptance authority	Only ROOTS may approve UAT results, exceptions and final acceptance

Appendix A — Internal Requirements Traceability Matrix

ROOTS project management and QA shall use this matrix to track procurement coverage, implementation status, evidence and acceptance. Any deviation requires a formally approved ROOTS change record.

Area	ID	Requirement	Implementation status	Evidence / decision reference
Scope and governance	GOV-001	Apply the sole-reference and conflict rules	Not Started / In Progress / Evidence Ready / Accepted	
	GOV-002	Implement only the approved baseline and ROOTS change records	Not Started / In Progress / Evidence Ready / Accepted	
	GOV-003	Provide weekly working-software demonstrations	Not Started / In Progress / Evidence Ready / Accepted	
	GOV-004	Maintain issue tracker, decision log and requirements traceability	Not Started / In Progress / Evidence Ready / Accepted	
Architecture and data	ARC-001	Complete Architecture Gate before material coding acceptance	Not Started / In Progress / Evidence Ready / Accepted	
	ARC-002	Use ROOTS-owned repository, accounts and three environments	Not Started / In Progress / Evidence Ready / Accepted	
	DB-001	Implement eight minimum logical application tables	Not Started / In Progress / Evidence Ready / Accepted	
	DB-002	Enable/test RLS on all protected tables and views	Not Started / In Progress / Evidence Ready / Accepted	
Assessment and intelligence	DB-003	Implement versioning, immutable history, consent and audit	Not Started / In Progress / Evidence Ready / Accepted	
	DB-004	Use approved HMAC research pseudonymization	Not Started / In Progress / Evidence Ready / Accepted	
	ASM-001	Implement Appendix C 73-question / 13-module dataset verbatim	Not Started / In Progress / Evidence Ready / Accepted	
	ASM-002	Implement save, resume, progress, navigation and error recovery	Not Started / In Progress / Evidence Ready / Accepted	
	ASM-003	Implement exact validation and N/A/missing rules	Not Started / In Progress / Evidence Ready / Accepted	
	SCR-001	Implement Appendix C Scoring Rules dataset only	Not Started / In Progress / Evidence Ready / Accepted	
	SCR-002	Implement seven domains and all approved derived outputs	Not Started / In Progress / Evidence Ready / Accepted	
	SCR-003	Implement configuration lifecycle and calculation trace	Not Started / In Progress / Evidence Ready / Accepted	
	SCR-004	Pass at least 30 Golden Test cases	Not Started / In Progress / Evidence Ready / Accepted	
	AI-001	Restrict LLM to governed narrative language	Not Started / In Progress / Evidence Ready / Accepted	
	AI-002	Implement output schema, prohibited-claim checks, retry and deterministic fallback	Not Started / In Progress / Evidence Ready / Accepted	

Area	ID	Requirement	Implementation status	Evidence / decision reference
	AI-003	Implement complete model/prompt/rules governance log	Not Started / In Progress / Evidence Ready / Accepted	
Report and website	RPT-001	Deliver interactive and PDF report with all 19 sections	Not Started / In Progress / Evidence Ready / Accepted	
	RPT-002	Use immutable canonical report JSON and approved visual reference	Not Started / In Progress / Evidence Ready / Accepted	
	WEB-001	Deliver all operational, informational and legal pages in Section 6	Not Started / In Progress / Evidence Ready / Accepted	
	WEB-002	Enforce analytics exclusion on sensitive routes	Not Started / In Progress / Evidence Ready / Accepted	
	ADM-001	Deliver Admin, configuration, audit and export functions	Not Started / In Progress / Evidence Ready / Accepted	
Security and delivery	SEC-001	Implement encryption, private storage, MFA, rate limits and session controls	Not Started / In Progress / Evidence Ready / Accepted	
	SEC-002	Provide processor/PHI data-flow map and required provider agreements/configuration	Not Started / In Progress / Evidence Ready / Accepted	
	SEC-003	Implement audit retention, incident notice, backups and restore	Not Started / In Progress / Evidence Ready / Accepted	
	QA-001	Deliver all required test suites and reports	Not Started / In Progress / Evidence Ready / Accepted	
	NFR-001	Meet performance, compatibility and accessibility targets	Not Started / In Progress / Evidence Ready / Accepted	
	DOC-001	Deliver complete documentation and SBOM	Not Started / In Progress / Evidence Ready / Accepted	
	HND-001	Complete repository/account transfer and recorded knowledge transfer	Not Started / In Progress / Evidence Ready / Accepted	
	SUP-001	Honor 90-day warranty and three-month complimentary support	Not Started / In Progress / Evidence Ready / Accepted	
Report and website	UI-001	Implement hash-locked C-05 screen specification and deliver responsive/accessibility/visual-regression evidence	Not Started / In Progress / Evidence Ready / Accepted	

Appendix B — Final Delivery Evidence Index

Evidence ID	Artifact	Location/version	ROOTS review
E-01	Repository and tagged release		
E-02	Staging and Production URLs		
E-03	Architecture/Data Flow/ERD		
E-04	OpenAPI specification		
E-05	Scoring release + Golden Test Pack		
E-06	Report JSON/PDF visual references		
E-07	Security/RLS/OWASP evidence		
E-08	Performance/accessibility/browser reports		
E-09	Backup/restore evidence		
E-10	SBOM/license/processor/cost registers		
E-11	Bug tracker and UAT report		
E-12	Handover recording and access checklist		
E-12	C-05 route inventory, component catalog, responsive screenshots, accessibility and visual-regression evidence		

Appendix C — Controlled Data Annex Register

The five files listed below are released, hash-locked and indivisible components of this Master Requirements document. Their exact filenames, versions, approval date and SHA-256 hashes identify the only executable content release. C-01 and C-02 are present and approved; assessment and scoring implementation may begin after checksum verification. No historical questionnaire, scoring example, sample report or website draft is authoritative.

Annex	Controlled dataset	Required release	Version / date / SHA-256	ROOTS approval record
C-01	Canonical Question Bank	Exactly 73 stable IDs across exactly 13 ordered modules; exact wording, options, types, validation and conditional logic	v1.0.0 / 21 July 2026 02_ROOTS_AI_C01_Canonical_Question_Bank_v1.0.0.xlsx SHA-256: 36785cc15e0be8a7781d7b4eec 51eb4be1ca31b712d622e24c07 1f8f66453802	ROOTS internal approval record: ROOTS-CR-2026-07-21-C-01
C-02	Canonical Scoring Rules	Every mapping, weight, reverse flag, threshold, formula constant, protective factor, driver, confidence rule and 30-case Golden Test Pack	v1.0.0 / 21 July 2026 03_ROOTS_AI_C02_Canonical_Scoring_Rules_and_Golden_Tests_v1.0.0.xlsx SHA-256: fe62560a0abc5a4da541516ad2 e8e20a1a6511f986d0c757f0841 14de530b226	ROOTS internal approval record: ROOTS-CR-2026-07-21-C-02
C-03	Report Content & Visual Reference Pack	Approved 19-section copy contract, templates, labels, safety rules and reference rendering	v1.0.0 / 21 July 2026 04_ROOTS_AI_C03_Report_Content_and_Visual_Reference_Pack_v1.0.0.pdf SHA-256: 877d71a7fd815e31a410325edc a7b9610216f1a50bab9b6944c3 5c5b9ce5256f	ROOTS internal approval record: ROOTS-CR-2026-07-21-C-03
C-04	Website Content & Legal Copy Pack	Approved content for all Phase 1 routes, privacy/terms/cookie/medical/AI notices and system messages	v1.0.0 / 21 July 2026 05_ROOTS_AI_C04_Website_Content_and_Legal_Copy_Pack_v1.0.0.docx SHA-256: b9f5ea308bb8f7b012a1805521f 722b9db5c6899b26dc18d82b74 d1498500bac	ROOTS internal approval record: ROOTS-CR-2026-07-21-C-04
C-05	UI/UX Screen Implementation Specification	Every Phase 1 screen, route, shell, navigation item, role visibility rule, component state, responsive behavior and UI acceptance artifact	v1.0.0 / 21 July 2026 07_ROOTS_AI_C05_UI_UX_Screen_Implementation_Specification_v1.0.0.pdf SHA-256: 450a74f6933dc1c11b18ba7efab 54e14257b8c190a3cd0d955080 57ac00cb731	ROOTS internal approval record: ROOTS-CR-2026-07-21-C-05

C.1 Canonical Question Bank Minimum Schema

Field	Mandatory rule
questionnaire_version	Immutable semantic version shared by all 73 records
module_id / module_order / module_title	Exactly 13 unique ordered modules; no duplicate order
question_id / question_order	Exactly Q1–Q73 or another explicitly approved stable 73-ID sequence; unique and gap-free
question_text	Exact approved participant-facing wording
question_type	Approved enum such as numeric, single select, multi-select, Likert, boolean, date or free text
answer_options	Stable option IDs, labels and stored values; no UI-label-as-logic dependency
required / allow_na	Explicit booleans; N/A is never inferred
validation	Range, length, unit and format rules plus approved error text
conditional_logic	Machine-readable trigger, operator, value and resulting show/hide/require action
scoring_eligible	Explicit boolean separating scored items from contextual items

C.2 Canonical Scoring Rules Minimum Schema

Field	Mandatory rule
scoring_version	Immutable semantic version linked to every score and report
question_id / domain_id	Exact relationship to one of MR, HS, SR, CH, SL, IB or BS, or explicit contextual-only status
option_points / normalization	Deterministic conversion from stored answer to normalized value
weight / reverse_scored	Explicit numeric weight and boolean reverse flag for every scored question
N/A / missing behavior	Exclude N/A; never impute; domain null below 50% eligible coverage unless changed in this annex
domain formula	Numerator, denominator, scale and rounding rule
derived formulas	Biological State, Opportunity and Recovery Potential formulas and rounding
classifications	Non-overlapping inclusive boundaries, labels and colors
drivers / evidence / confidence	Deterministic thresholds, tie rules, coverage rules and outputs
test vectors	Input and exact expected output for every boundary and reverse-scored branch

Appendix D — Consolidated Requirements Decisions

The decisions below are final for Version 3.3 and may be changed only by a formally approved ROOTS change record. Historical alternatives are discarded.

Subject	Approved Version 3.3 decision	Required evidence
Technical stack	Next.js/React and server-side TypeScript strict mode; Supabase; Vercel; Puppeteer; OpenAI only for governed narrative	Architecture Gate and deployed environments
Database scope	At least eight logical application tables plus managed auth, views, buckets and optional normalized tables	ERD, migrations and table inventory
RLS	Enabled and negatively tested on every protected table and view	Automated RLS test report
Assessment	Exactly 73 questions across exactly 13 modules from hash-locked Annex C-01	Automated count/schema check and UAT
N/A and missing	N/A excluded; missing never imputed; domain null below approved coverage threshold	Golden tests
Scoring	Deterministic only; no demographic adjustment or invented rule outside Annex C-02	Trace output and Golden Test Pack
Website	All 18 operational, informational, legal and admin routes listed in Section 6	Route inventory and page UAT
Report	Interactive and PDF report contain all 19 required sections	Canonical JSON, reference PDF and visual tests
AI role	Language renderer only; cannot calculate or alter scores, drivers, confidence or eligibility	Governance and fallback tests
Performance	Exact measurable budgets in Section 12; not best-effort targets	Representative-load performance report
Security/privacy	Private storage, MFA for privileged roles, processor controls, audit, consent, retention, backup and restore	Security, processor and restore evidence
Analytics	No analytics, replay or pixels on assessment, report, auth or admin routes	Network inspection and configuration evidence
Repository/accounts	ROOTS-AI™ ownership and control from Day 0	Owner screenshots/access register
Timeline	Thirty calendar days from Effective Date, including seven-day UAT; complete Release Candidate by Day 23 and Final Acceptance by Day 30; payment follows accepted evidence	Approved milestone certificates
Warranty/support	90-day defect warranty plus three calendar months complimentary support from Final Acceptance	Support log and SLA evidence

END OF DOCUMENT

ROOTS-AI™ — CONTROLLED CORRECTION SHEET

Master Product & Implementation Requirements v3.3.1

This sheet is an integral part of v3.3 and supersedes only the clauses expressly identified below. All other requirements remain unchanged.

1. Section 6.1 — Design and Content System

Superseded text: the fixed “Quiet Luxury” aesthetic prescription and any interpretation that treats the listed palette or Inter typography as the final visual benchmark.

Controlling replacement: Functional, accessibility, responsive, content-governance and approved naming requirements in Section 6.1 remain mandatory. Final PUB-01 visual styling, artistic direction, component appearance, design quality, typography treatment, visual hierarchy and Golden Screen acceptance are governed by **ROOTS_AI_Visual_Direction_and_Golden_Screen_Acceptance_Brief_v1.0_FINAL**. Existing SVGs remain functional/content wireframes and do not constitute the final aesthetic design. A Golden Screen becomes the controlling visual-system reference only after written ROOTS-AI approval.

The approved visual-direction brief studies quality principles from Oura, Function Health, WHOOP and Apple Health without copying their layouts, assets, charts, trade dress or proprietary interaction patterns. The final expression must remain recognisably ROOTS-AI™.

2. Section 15.2 — G2 / Milestone Interpretation

Clarification required to remove the G2/M1–M2 ambiguity: the Day-7 G2 row describes the intended early delivery trajectory for the Assessment/Data workstream; it does not override the signed commercial milestone allocation.

For payment and formal milestone acceptance: M1 requires the assessment shell plus autosave/resume foundation, together with the applicable G0–G2 foundation evidence (repository/environment control, architecture, schema/migrations, authentication, RLS foundation, traceability/test evidence and Staging demonstration). M2 requires completion of the canonical 73-question / 13-module assessment, validation/progress, deterministic seven-domain scoring, versioning and all 30 Golden Tests. A delay of full 73-question completion beyond Day 7 does not by itself convert that M2 deliverable into an M1 payment condition; however, the M1 shell/autosave/resume foundation remains mandatory for M1 acceptance.

3. Appendix C — Controlled Data Annexes

Appendix C remains authoritative and is **not** converted from executable source formats to PDF. The final vendor package must include the exact hash-locked source files identified by Appendix C wherever an executable/editable source is specified.

Annex	Required source	Verified SHA-256 status
C-01	02_ROOTS_AI_C01_Canonical_Question_Bank_v1.0.0.xlsx	MATCHES Appendix C
C-02	03_ROOTS_AI_C02_Canonical_Scoring_Rules_and_Golden_Tests_v1.0.0.xlsx	MATCHES Appendix C
C-04	05_ROOTS_AI_C04_Website_Content_and_Legal_Copy_Pack_v1.0.0.docx	MATCHES Appendix C

4. Precedence

If the original v3.3 wording conflicts with this correction sheet on the three matters above, this v3.3.1 correction controls. Nothing in this correction relaxes scientific, security, privacy, legal, accessibility, deterministic-scoring, traceability, testing or handover requirements.